

Workshop #1: Welcome to the R environment

What is R and how
do we open it, why would
we use it, and how?

About Me- Sam Swift, PhD, MPH

- Emergency Medical Technician
- UNM MPH
- NMDOH Epidemiologist
- University of Miami PhD (Social Epidemiology)
- UNM Postdoc
- Causal Inference Enthusiast
- SAS & R, some Stata, ArcGIS, & Mplus
- UNM Assistant Professor



Questions about the Workshop???

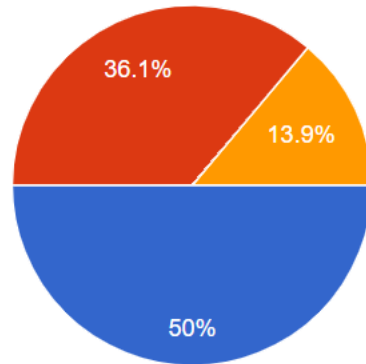
Email me!!!

samswift@salud.unm.edu

Survey Results (I know pie charts are bad)

How much experience do you have with R programming?

36 responses

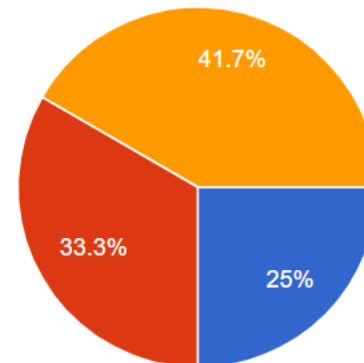


● No experience
● Some experience
● Lots of experience

25% have no coding experience at all

How much experience do you have with other types of statistical programming (SAS/Stata)?

36 responses



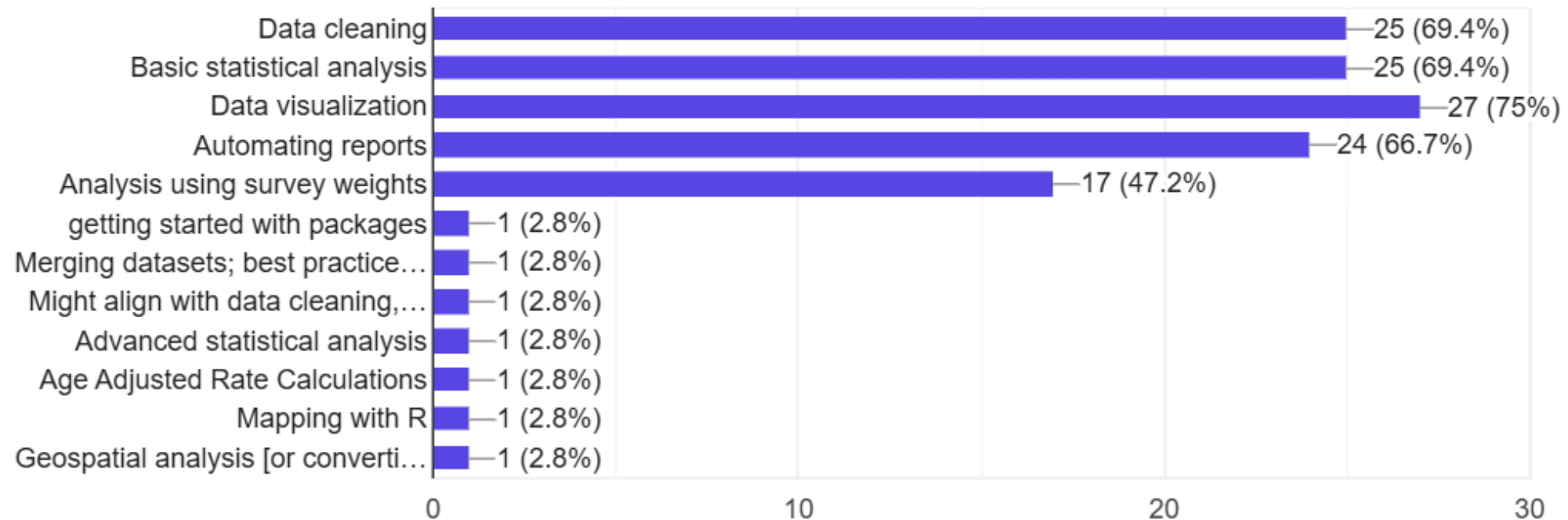
● No experience
● Some experience
● Lots of experience

Today may be basic for the lots of experience people, but we can ramp up

More survey results

What topics would be most useful for you to learn? Check all that apply

36 responses



Final (select) survey results

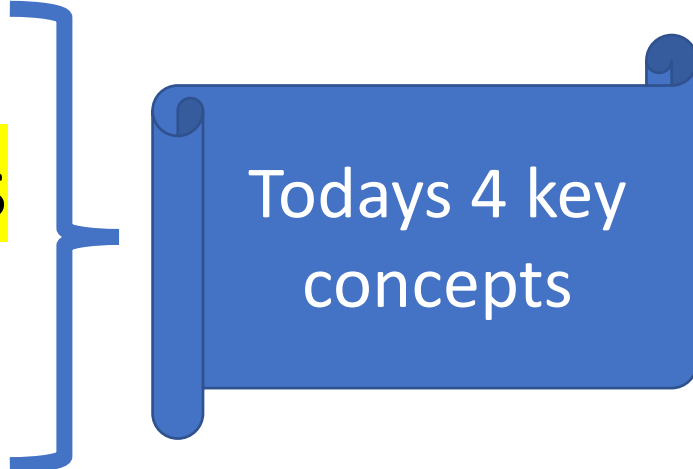
- What should I cover?
 - Basic: Syntax, data viz- merges
 - More advanced: github, R shiny, functions similar to macros, mapping, multiple imputation
- What helps you learn?
 - Instructor having empathy
 - Not too much info
 - Hands on/ interactive work/ homework
 - **STOP DURING LECTURE FOR Qs**



Presentation contents

What is R/ R studio

- Objects
- Functions
- Scripts
- Packages
- Paths
- Comments



Today's 4 key concepts

Learning objectives

- Be able to save data on your computer and read data into R studio
- Understand the definitions of:
 - Objects
 - Functions
 - Scripts
 - Packages
 - File paths
 - Comments

Workshops outline

- Session 1. (August 4th 2026): R basics, including what each window pane means, different menus, syntax, and packages
- Session 2. (August 11th 2026): R basics review, introduction to data cleaning and variable creation
- Session 3. (August 19th 2026): Intermediate data cleaning, joining datasets, and basic statistical analysis in R
- Session 4. (August 25th 2026): Data visualization with ggplot2
- Session 5. (September 1st 2026): Automating reports with R markdown and integration with GitHub
 - Survey weights here as well?
 - Loops?

Class Repository

- https://github.com/SamSwift505/NMDOH_R_WORKSHOPS_26
- All materials will be available here
- Later we do a little more with github



Installing R and R studio

- We need R and R studio to do this!
- Instructions on how to install both R and R studio are below points A1-A3 here:
- <https://rstudio-education.github.io/hopr/starting.html>Links to an external site.
- A video on installing R and R studio
- <https://youtu.be/Tb3R4GLJ45U>

To run R Studio on your computer, you need to have both R and R Studio installed.

- Both of these are applications that you download
- You need admin privileges!

Entertaining meme

If statistics programs/languages were cars...





**Language &
Environment**

Statistics



FREE!

What is
 **?**

Community



Customizable

Online Training



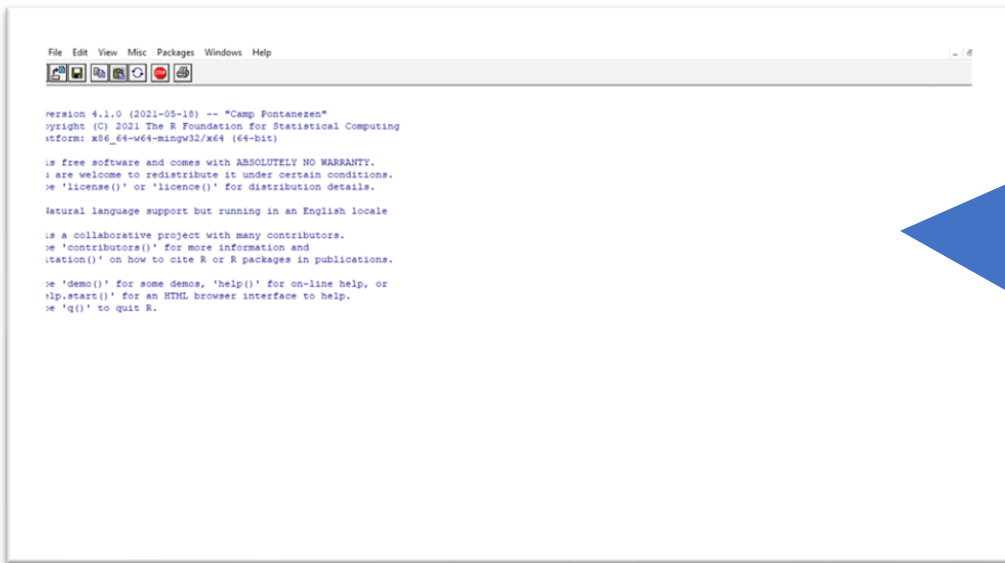
There are Some Disadvantages to R/ R studio

- Can be less intuitive to learn than other programs
- Updates require maintenance
- Security
- Can be slower than other languages
- Some packages are not great quality
- Memory management



R vs R studio- Why do we need both?

- By installing R, you install the code that R studio runs on
- R studio us a “user friendly” (?) interface that we use to code in R



```
File Edit View Misc Packages Windows Help
[Icons]

version 4.1.0 (2021-05-18) -- "Camp Fontanezen"
copyright (C) 2021 The R Foundation for Statistical Computing
xform: x86_64-mingw32/x64 (64-bit)

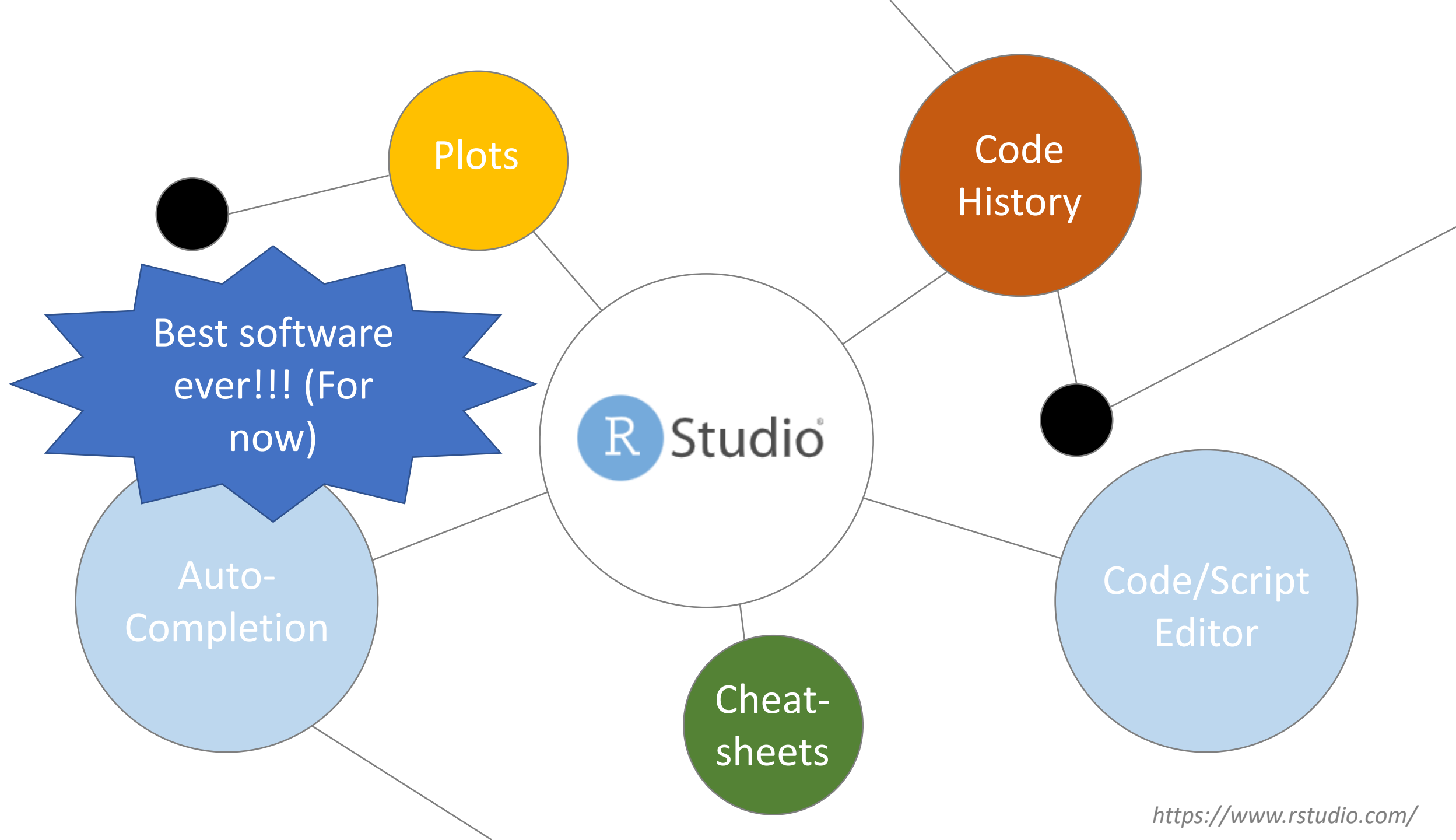
is free software and comes with ABSOLUTELY NO WARRANTY.
you are welcome to redistribute it under certain conditions.
see 'license()' or 'licence()' for distribution details.

Natural language support but running in an English locale

is a collaborative project with many contributors.
see 'contributors()' for more information and
'citation()' on how to cite R or R packages in publications.

see 'demo()' for some demos, 'help()' for on-line help, or
'help.start()' for an HTML browser interface to help.
see 'q()' to quit R.
```

It is possible to work directly in the R Console, but it is clunky!



RStudio

File Edit Code View Plots Session Build Debug Profile Tools Help

Go to file/function

Addins

Untitled1 x

Source on Save

Run

Environment History Connections Tutorial

Import Dataset 99 MiB

R Global Environment

Environment is empty

Help Viewer Presentation

Export

1:1 (Top Level) R Script

Console Terminal Background Jobs

R 4.5.0 ~

R version 4.5.0 (2025-04-11 ucrt) -- "How About a Twenty-Six"
Copyright (C) 2025 The R Foundation for Statistical Computing
Platform: x86_64-w64-mingw32/x64

R is free software; you are welcome to redistribute it under certain conditions.
Type 'license()' or 'licence()' for distribution details.

Natural language processing

R is a collaborative project. Type 'contributor()' or 'contributors()' for more details.
Type 'demo()' for some demos. Type 'help.start()' for an HTML browser interface to help.
Type 'q()' to quit R.

[Workspace loaded from ~/.RData]

> |

The script panel is where permanent programs can be saved

Objects you create will be saved here to the "global environment"

You can also freely code in the console here

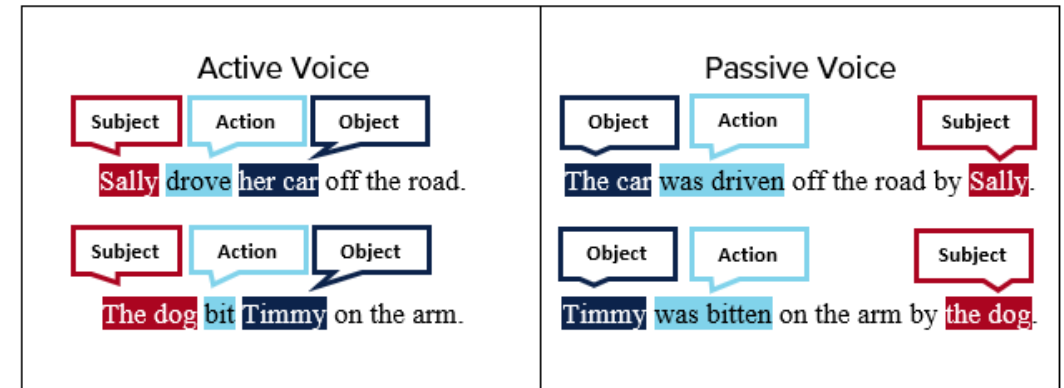
Help/ packages/Plots will appear here, also you can access packages and the help window from here

Code has syntax

- Think of lines of code like learning a new language
- To teach you this I do a special segment called

Syntax Breakdown

- By the end of this class you will know some basic words and structure, I do not expect you to speak the language
- Also please don't use passive voice

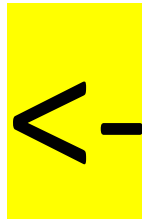


Key Terms

- **Objects** - Everything you store in R - datasets, variables, a list of village names, a total population number, even outputs such as graphs - are objects which are assigned a name and can be referenced in later commands.
- **Functions** - A function is a code operation that accept inputs and returns a transformed output.
- **Packages** - An R package is a shareable bundle of functions.
- **Scripts** - A script is the document file that hold your commands.

Objects

- Objects are things you make, like datasets, tables, graphs and more
- The arrow operator (less than, dash) is what we use to assign things into objects
- That operator looks like this

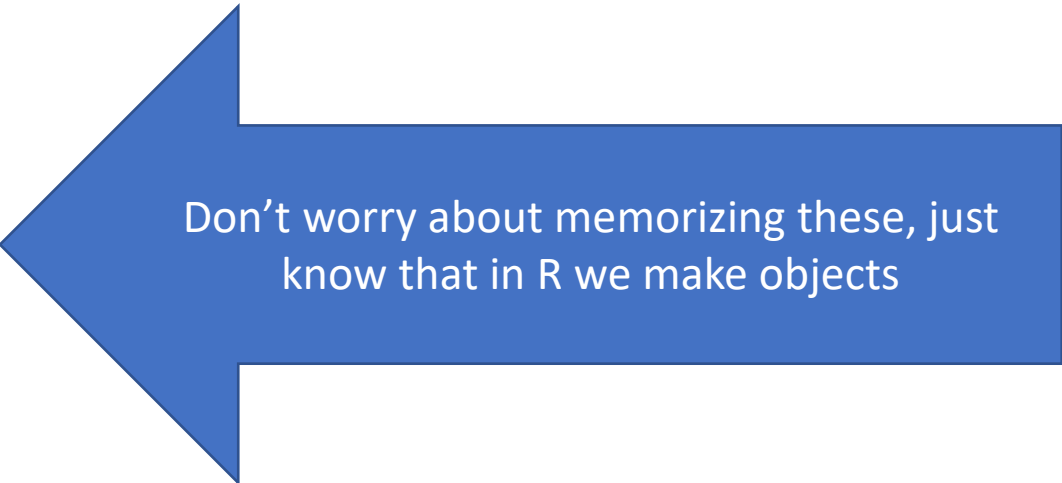


The syntax we will use to make objects will look like this:

object_name <- value

Objects Continued

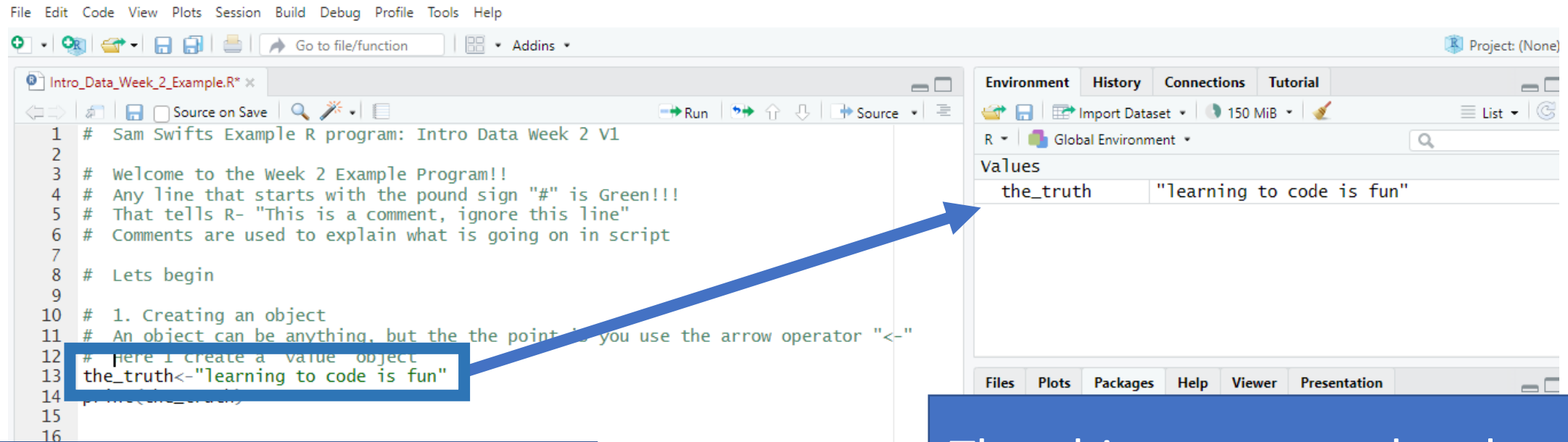
- There are lots of different types of objects
 - We will mostly use data frames (datasets)
 - And plots (graph/plots)
 - Today I will show a simple “value” object
- There are also many other more advanced ones
 - Vectors
 - Matrices
 - Lists (data frames are special lists)
 - Arrays



Don't worry about memorizing these, just know that in R we make objects

Objects Continued

- When you make an object, it will appear in the top right “environment window”



The screenshot shows the RStudio interface. The main editor window displays a script titled 'Intro_Data_Week_2_Example.R'. The script contains several lines of R code, including comments and a line that creates an object: `the_truth <- "learning to code is fun"`. This line is highlighted with a blue box. A blue arrow points from this line to the Environment window on the right. The Environment window shows the current environment (Global Environment) and lists the objects created. Under the 'Values' section, the object 'the_truth' is listed with its value 'learning to code is fun'.

```
1 # Sam Swifts Example R program: Intro Data Week 2 V1
2
3 # Welcome to the Week 2 Example Program!!
4 # Any line that starts with the pound sign "#" is Green!!!
5 # That tells R- "This is a comment, ignore this line"
6 # Comments are used to explain what is going on in script
7
8 # Lets begin
9
10 # 1. Creating an object
11 # An object can be anything, but the the point is you use the arrow operator "<-"
12 # here I create a value object
13 the_truth <- "learning to code is fun"
14 print(the_truth)
15
16
```

Environment | History | Connections | Tutorial

R | Global Environment

Values

the_truth	"learning to code is fun"
-----------	---------------------------

Files | Plots | Packages | Help | Viewer | Presentation

I ran this code

The object appeared up here in “Environment”

Syntax Breakdown

Break it down

the_truth<-"learning to code is fun"

Like parenthesis, if you open quotes you must close them

The object we created, can be named anything AS LONG AS THERE ARE NO SPACES

A string of text- the computer knows its text because its in quotes

Our friend the arrow operator, says take what follows and put it in the object

Error messages

- If you make a mistake, R will error out!!!
- Usually this happens with angry red text in the console window, or an X on the line where the error is

```
28  
29 the_truth<-(learning to code is fun  
30  
31
```

```
31 # lets try to print an object that we haven't made  
32 # this will also cause an error  
33 print(object_that_does_not_exist)
```

33:29 (Top Level) ↕

Console

Terminal ×

Background Jobs ×

R • R 4.5.0 • ~/

```
> print(object_that_does_not_exist)  
Error: object 'object_that_does_not_exist' not found  
> |
```

Errors are
your friends—
they help you
learn!!

Debugging Errors

- Our class is a community of practice
- Lets work together in class discussions to help
- Do not be embarrassed
- If no one helps I will help
- You can also paste errors into google (which I will demonstrate)
- You can also ask AI but its often wrong

Functions

- Functions perform tasks and operations
- A function often wraps around an object, using parenthesis
- Example:

Break it down

```
30  
37 # Create a value that is 4  
38 four<-4  
39 # Take the square root of that value  
40 sqrt(four)  
41  
37:1 (Top Level) ↕  
Console Terminal × Background Jobs ×  
R • R 4.5.0 • ~/   
> # Create a value that is 4  
> four<-4  
> # Take the square root of that value  
> sqrt(four)  
[1] 2  
> |
```

Square root of 4 is 2

sqrt(four)

An object I made that is the number 4

The function– Note the parenthesis wrapping the object

Packages

- From your reading:
- “An R package is a shareable bundle of code and documentation that contains pre-defined functions.”
- This is the most exciting thing about R!!
- In this class we will use three:
 - tidyverse (next week)
 - dplyr (next week)
 - ggplot2



Ok... so what is a package??

- Basically installing a package is like downloading more functions within R studio
- Once you install ggplot2, you get access to a lot of amazing data visualization functions
- Once you download it, its there forever, but you need to ask R studio to load it in each session (shown in 2 slides)

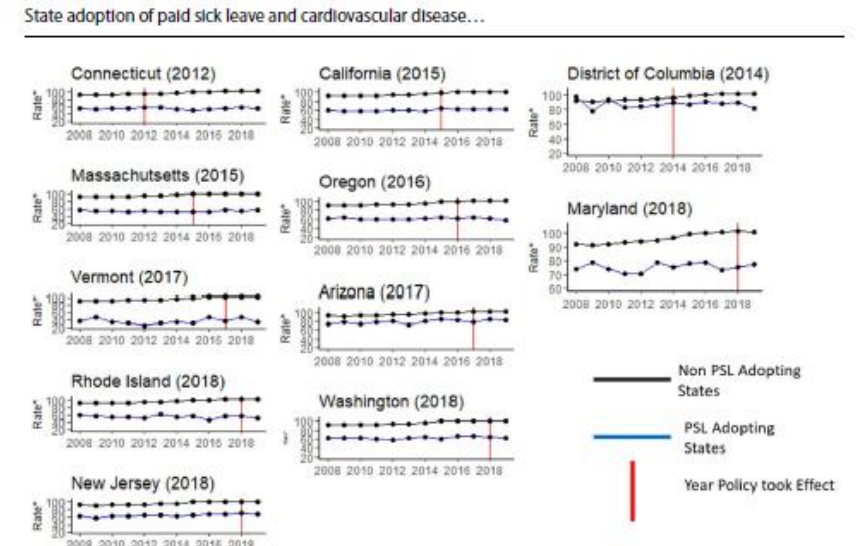
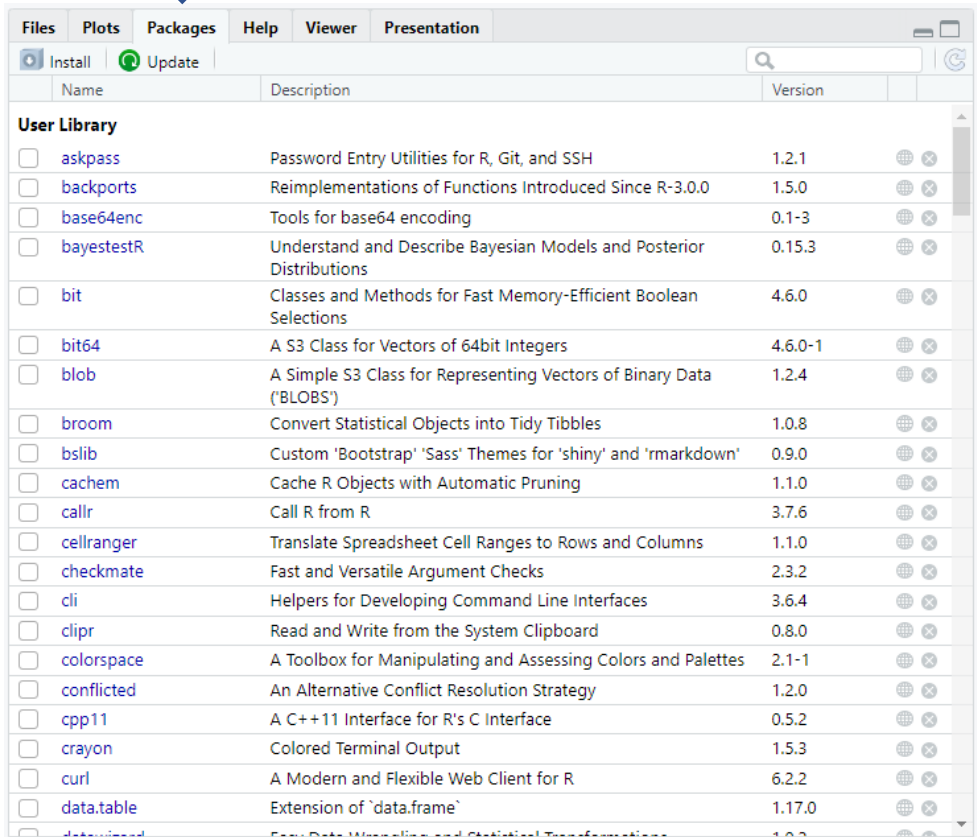


Fig.2 Annual cardiovascular disease mortality rate among persons 15-64 in states that adopted State-level paid sick leave policies (treated) vs non-adopting (NA) states (untreated), 2008 to 2019. Rates are county-level mortality rates due to cardiovascular disease among persons 15-64, per 100,000 persons. Non-adopting states (as of 2019) in the Northeast are Maine, New Hampshire, Pennsylvania, and New York. Non-adopting states (as of 2019) in West are Colorado, Idaho, New Mexico, Montana, Utah, Nevada, Wyoming, Alaska, and Hawaii. Non-adopting states (as of 2019) in the South are Delaware, Florida, Georgia, North Carolina, South Carolina, Virginia, West Virginia, Alabama, Kentucky, Mississippi, Tennessee, Arkansas, Louisiana, Oklahoma, and Texas

Downloading Packages

In the “packages” tab you can search for and download packages



You install it once using `install.packages` function

- You can also type:
`install.packages("<package name>")`
into the console

- Once the package is downloaded, the contents can be made available by using:
`library(<package name>)`

You call the package in using the `library` function in every time

SYNTAX BREAKDOWNNNNN

Function to install the first time

```
install.packages("tidyverse")  
# here is the library function  
library(tidyverse)
```

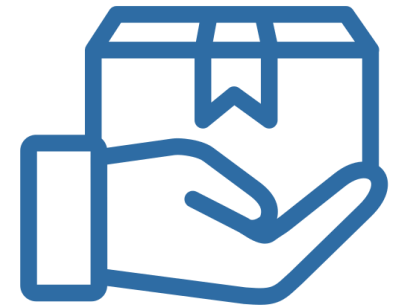
This is a comment, a line that does nothing but is used for documentation/ notes (starts with a hashtag)

Function to read package in every time- you must do this at the beginning of every R session

The package we are installing/ using

Commonly Used Packages

- **tidyverse**: A collection of packages designed for data manipulation
- **dplyr**: Shortcuts for sub-setting, summarizing, rearranging, and joining data
- **tidyr**: Changing the layout of the data sets
- **stringr**: Tools for regular expressions and character strings
- **lubridate**: Makes working with dates and times easier
- **ggplot2**: Create graphics
- **R Markdown**: Can be used for automated reporting. Reports can be exported as HTML, pdf, or MS Word document

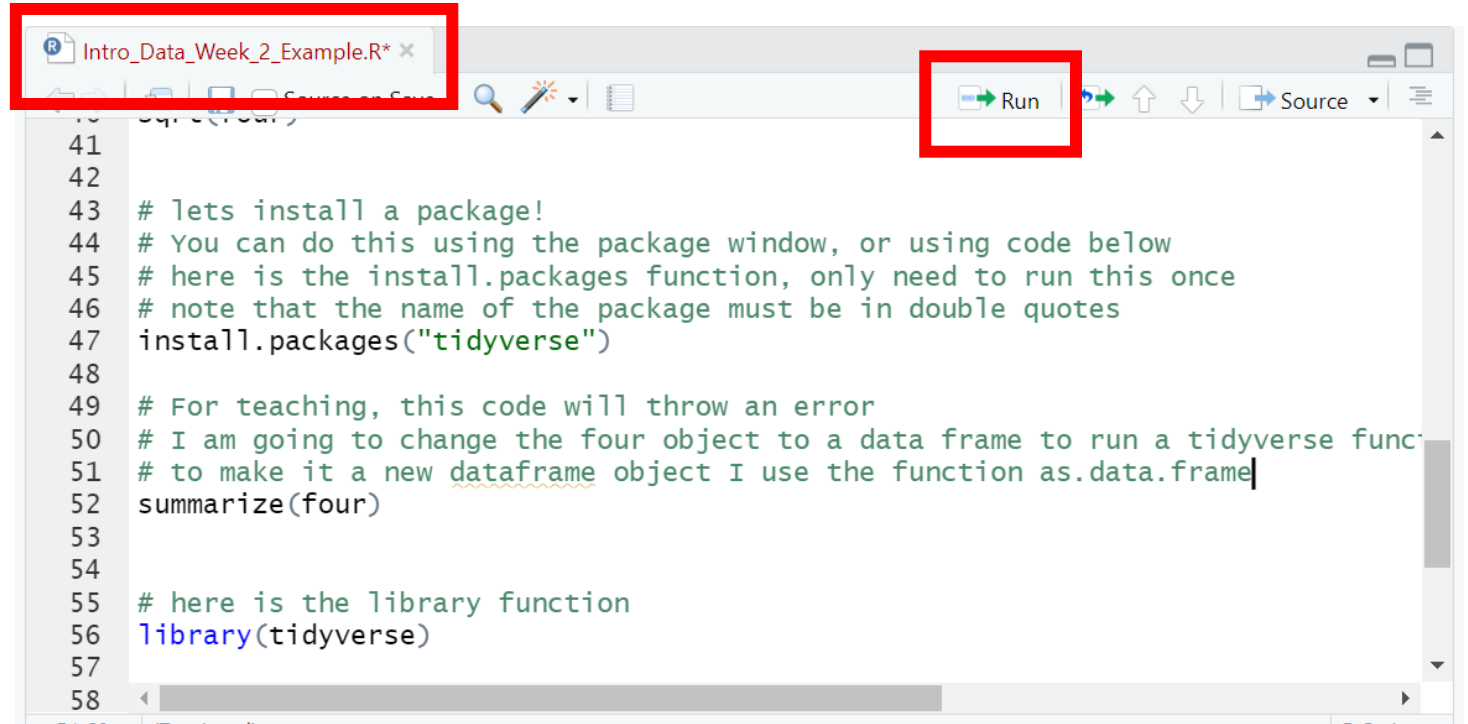


Scripts

- By coding in the “scripts” window you have a file that you can come back to in later sessions
 - All statistical software has something like this
 - You will be required to save an upload a script for the homework

Red text means your version is unsaved

The run button will run highlighted code



The screenshot shows an R script editor window titled 'Intro_Data_Week_2_Example.R*'. The window has a toolbar with a 'Run' button (a green arrow) highlighted by a red box. The script content is as follows:

```
41  
42  
43 # lets install a package!  
44 # You can do this using the package window, or using code below  
45 # here is the install.packages function, only need to run this once  
46 # note that the name of the package must be in double quotes  
47 install.packages("tidyverse")  
48  
49 # For teaching, this code will throw an error  
50 # I am going to change the four object to a data frame to run a tidyverse func  
51 # to make it a new dataframe object I use the function as.data.frame|  
52 summarize(four)  
53  
54  
55 # here is the library function  
56 library(tidyverse)  
57  
58
```

The file name 'Intro_Data_Week_2_Example.R*' is highlighted by a red box, indicating it is unsaved. The 'Run' button in the toolbar is also highlighted by a red box.

File paths and reading in data

- One of the hardest things for new users of statistical software is reading in data
- For this workshop, there are three broad steps
 1. Download the data from GitHub (demonstrated in video today)
 2. Save the data in an organized place on your computer
 3. Use code to tell R studio where the data is and read it in as a new object.

File paths are a key concept for the week- we use these to read data in and out

Downloading the data from github

NMDOH_R_WORKSHOPS_26 / Week_1 / Week_1_data.csv

SamSwift505 Rename Week_1_data.csv to Week_1/Week_1_data.csv 29e744e · 2 months ago History

Preview Code Blame 34 lines (34 loc) · 1.52 KB Raw

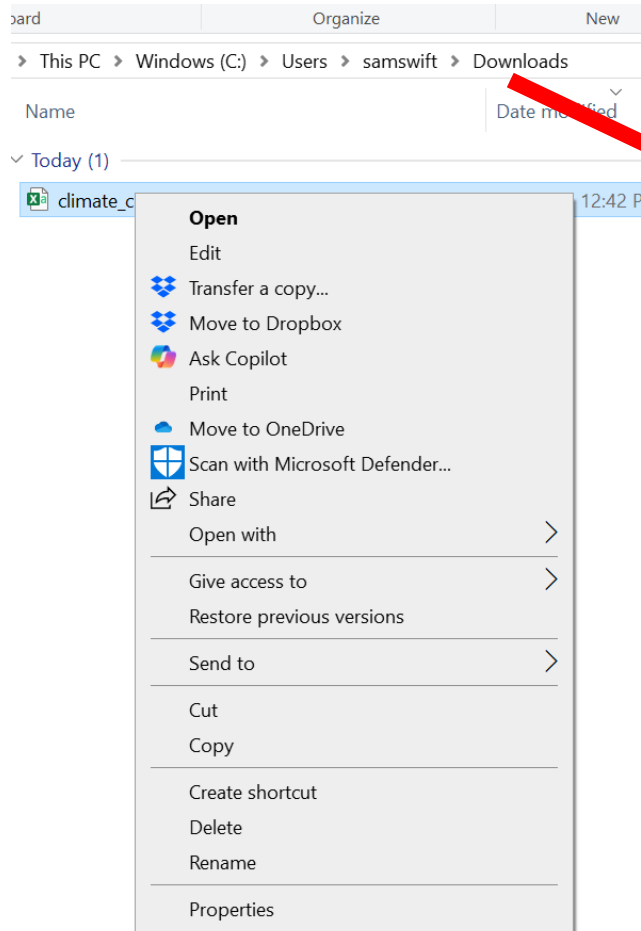
Search this file

1	County	Child_pov	Med_inc	Checkup_adult_per	Flu_18_per	County_designation	Cancer_deaths
2	Bernalillo County	0.24	56920	0.63	0.48	metro	1,147
	-	-	-	-	-	-	-

Double clicking this downloads
your data

Save the file somewhere organized

There are different ways to save the downloaded folder, which I will show on the video

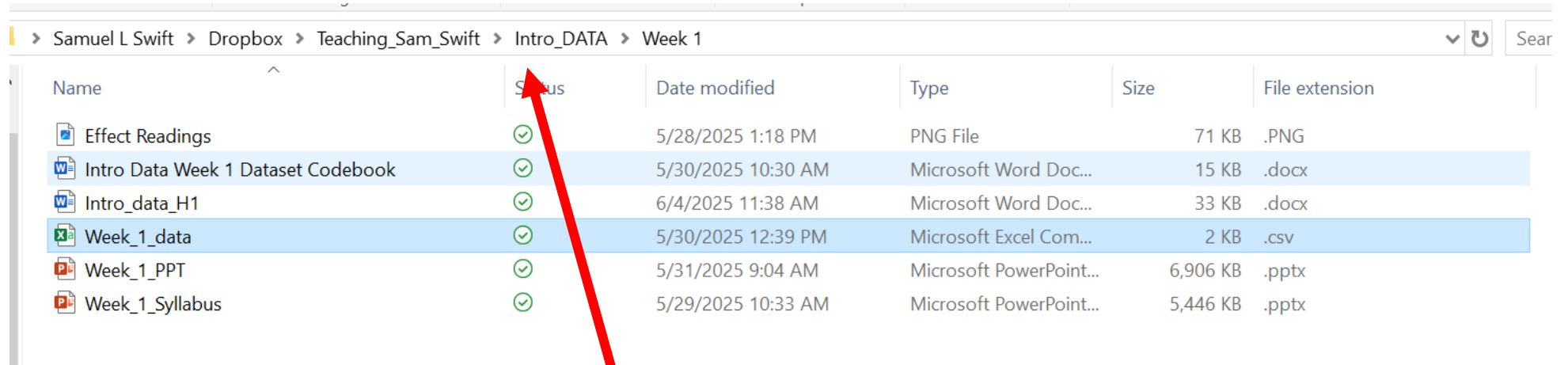


> Dropbox > Teaching_Sam_Swift > Intro_DATA > Class_datasets

Name	Status	Date modified	Type	Size
climate_change	✓	6/6/2025 9:28 AM	Microsoft Excel Com...	4 KB
climate_change	✓	6/6/2025 11:24 AM	R File	3 KB
PReP	✓	6/6/2025 11:59 AM	Microsoft Excel Com...	5 KB
PReP	✓	6/6/2025 11:58 AM	R File	2 KB
SNAP_MCH	✓	6/6/2025 11:18 AM	Microsoft Excel Com...	4 KB
SNAP_MCH	✓	6/6/2025 11:22 AM	R File	3 KB

WHERE YOU SAVE IT IS THE FILEPATH

Looks like this in
windows file
explorer



File Explorer path: > Samuel L Swift > Dropbox > Teaching_Sam_Swift > Intro_DATA > Week 1

Name	Status	Date modified	Type	Size	File extension
Effect Readings	✓	5/28/2025 1:18 PM	PNG File	71 KB	.PNG
Intro Data Week 1 Dataset Codebook	✓	5/30/2025 10:30 AM	Microsoft Word Doc...	15 KB	.docx
Intro_data_H1	✓	6/4/2025 11:38 AM	Microsoft Word Doc...	33 KB	.docx
Week_1_data	✓	5/30/2025 12:39 PM	Microsoft Excel Com...	2 KB	.csv
Week_1_PPT	✓	5/31/2025 9:04 AM	Microsoft PowerPoint...	6,906 KB	.pptx
Week_1_Syllabus	✓	5/29/2025 10:33 AM	Microsoft PowerPoint...	5,446 KB	.pptx

Looks like this in
R studio

```
# Lets learn how to read in data  
# TO MAKE THE LINE BELOW WORK YOU MUST CHANGE THE PATH- KEY CONCEPT FOR THE WEEK  
# The path is where the data are, slashes are subfolders |  
data_week_1<-read.csv("C:/Users/samswift/Dropbox/Teaching_Sam_Swift/Intro_DATA/Week 1/week_1_data.csv")
```

File extensions

- In general, they tell us a lot of stuff about what type of file we are using
- In datasets, they can tell us about the statistical software we are using,

- The datasets file extensions also tell us the delimiter

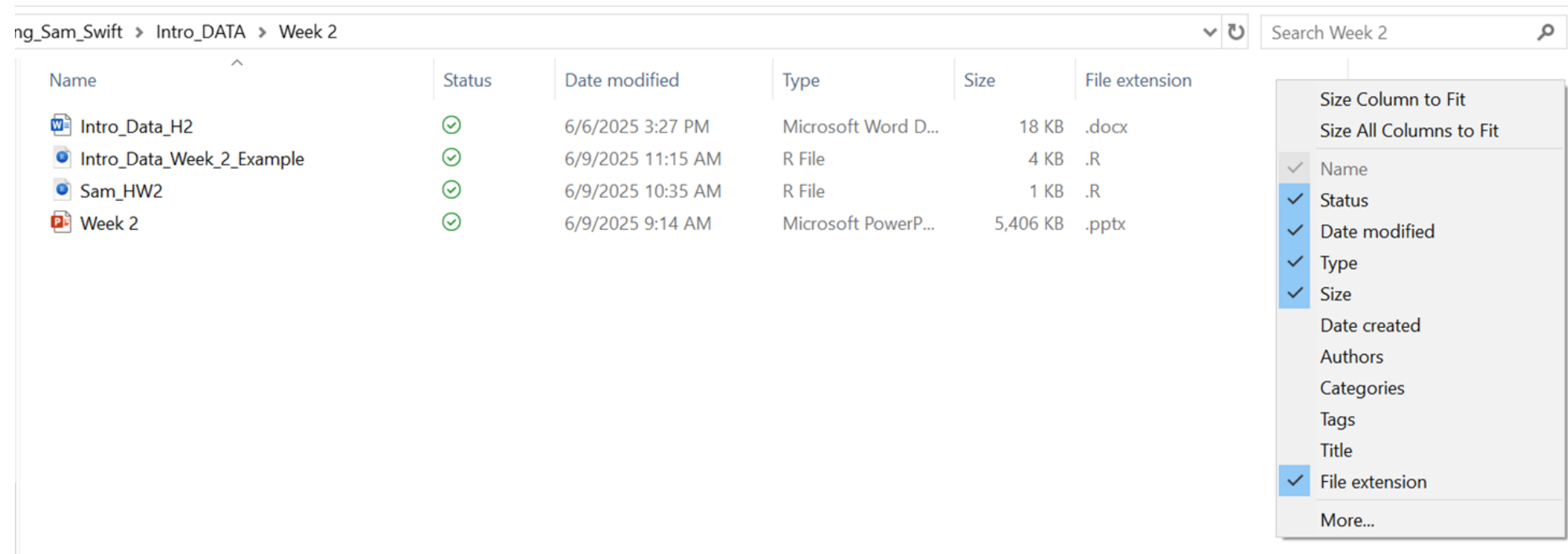
- R packages can read in data from any statistical software



Add extensions to windows explorer/ finder

Windows explorer: <https://lazyadmin.nl/win-11/show-file-extension-windows-11/>

Finder (Mac): <https://support.apple.com/guide/mac-help/show-or-hide-filename-extensions-on-mac-mchlp2304/mac>



Anyway, Delimiters

Delimiters: Delimiters are characters that mark the boundaries between individual pieces of data within a file.

In this class we use CSV because the data are simple and do not cause problems for learners

.csv: Indicates comma-separated values.

.txt: Indicates a general text file, which can also be a delimited file.

.tsv: Indicates tab-separated values.

.psv: Indicates pipe-separated values.

CSV behind the scenes

What looks like this in Excel

	A	B	C	D	E	
1	age	wildfiresm	heatdays	asthma	sbp	hear
2	40	No	46	Yes	121.4	No
3	60	No	26	No	127.3	No
4	23	No	2	No	117	No
5	76	Yes	53	No	144.8	Yes
6	51	No	0	No	109.5	No
7	59	No	23	No	111.2	Yes
8	52	No	0	Yes	122.2	Yes
9	23	Yes	15	Yes	107	No
10	34	No	60	No	143.9	No
11	78	No	60	No	119.1	No
12	66	No	23	No	108.7	No
13	18	No	11	Yes	116.6	Yes

```
"age","wildfiresmoke","heatdays","asthma","sbp","headaches","anxiety"  
40,"No",46,"Yes",121.4,"No","No"  
60,"No",26,"No",127.3,"No","No"  
23,"No",2,"No",117,"No","No"  
76,"Yes",53,"No",144.8,"Yes","Yes"  
51,"No",0,"No",109.5,"No","No"  
59,"No",23,"No",111.2,"Yes","No"  
52,"No",0,"Yes",122.2,"Yes","No"  
23,"Yes",15,"Yes",107,"No","Yes"
```

I will only use CSV
in this class, and
this is why

Looks like this in its raw form, which
is easy for different software to
understand

Note the commas!

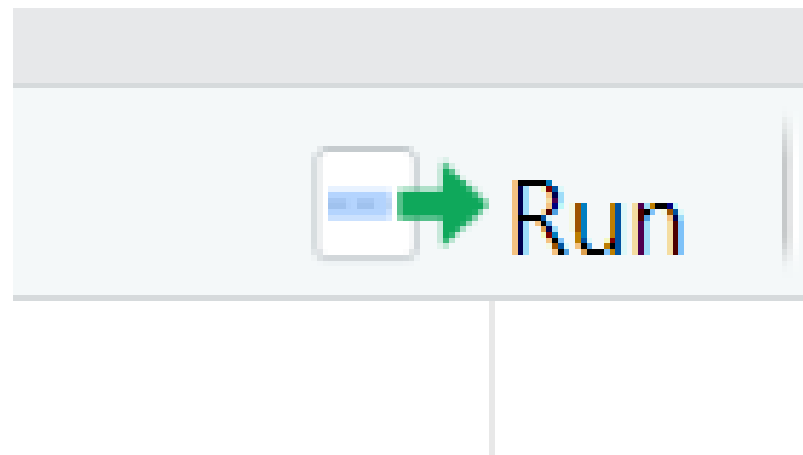
Comments

- Comments start with this symbol #
 - (AKA octothorpe, pound sign, hashtag)
- These are used to document your code
- If you highlight these and run them nothing happens
- These are extremely useful for
 - Reproducible science
 - Trying to be a good teacher teaching people to code for first time
- You will type your HW answers as comments this week

```
55 # here is the library function
56 library(tidyverse)
57
58 # Lets learn how to read in data
59 # TO MAKE THE LINE BELOW WORK YOU MUST CHANGE THE PATH- K
60 # The path is where the data are, slashes are subfolders
61
62 data_week_1<-read.csv("C:/Users/samswift/Dropbox/Teaching
```


Running Code

- To run code, you can highlight a line of code and push the run button
- If there are open parenthesis on two lines, you need to highlight both
- You can also click on a line and push control+enter
- Command+enter on Mac



They made
this button
way too
small!!!

Open Source: A Blessing and A Curse

- It is a scientific innovation that anyone, anywhere can write packages
- This increases reproducibility, shared knowledge, and innovation
- It also means sometimes syntax may vary slightly
- In extreme cases, packages can be wrong (none that we will be working with)



Questions?? Lets go to R

- EMAIL ME MORE QUESTIONS:

- samswift@salud.unm.edu



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